

**LEGALVISION: A KNOWLEDGE-DRIVEN AI
FRAMEWORK FOR LEGAL UNDERSTANDING AND
TRUST ASSESSMENT**

Project Id : 25-26J-127

Project Proposal Report

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B.Sc. (Hons) Degree in Information Technology
Specialized in Data Science

Department of Information Technology

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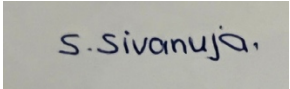
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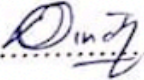
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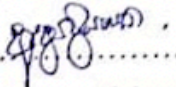
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ABSTRACT

Legal document analysis in property law remains complex and inaccessible, often requiring significant time and expertise to interpret deeds, leases, and partition agreements. Existing AI-powered legal tools provide surface-level insights but often lack transparency, explainability, and traceability, limiting their adoption in high-stakes legal contexts. Conventional models act as “black boxes,” generating conclusions without clear reasoning, leaving professionals unable to validate or trust the outputs.

This proposal introduces an explainable legal reasoning module that integrates prompt chaining, argumentation graphs, and large language models (LLMs) to generate step-by-step, auditable reasoning chains for Sri Lankan property law. The system links each insight to specific statutes, case precedents, and document clauses, providing traceability and defensibility. Additionally, logic trees and visual reasoning graphs enhance interpretability for both legal professionals and non-expert users, bridging the gap between advanced AI processing and practical usability.

Through innovations in structured reasoning, transparency, and visual interpretability, this project advances AI-powered legal assistance beyond automated summarization toward trustworthy, explainable, and legally accountable solutions. The outcome is a property law reasoning module that not only analyzes and explains decisions but also builds confidence in AI-supported legal processes, enabling better decision-making and risk assessment.

Subject Descriptors (ACM CCS)

- Computing methodologies → Artificial intelligence → Natural language processing → Reasoning systems
- Applied computing → Law, social and behavioral sciences → Legal informatics

Keywords

Property law, Explainable AI, Prompt chaining, Argumentation graphs, Legal reasoning, Trustworthy AI.

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List of Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
DB	Database
DL	Deep Learning
FAISS	Facebook AI Similarity Search
FastAPI	Fast Application Programming Interface (Python framework)
GDPR	General Data Protection Regulation
GPU	Graphics Processing Unit
HTTP/HTTPS	Hypertext Transfer Protocol / Secure Hypertext Transfer Protocol
JSON	JavaScript Object Notation
JWT	JSON Web Token
KG	Knowledge Graph
LLM	Large Language Model
LoRA	Low-Rank Adaptation
NER	Named Entity Recognition
NLP	Natural Language Processing
OCR	Optical Character Recognition
OIDC	OpenID Connect
PEFT	Parameter-Efficient Fine-Tuning
QoS	Quality of Service
RAG	Retrieval-Augmented Generation
RAD	Rapid Application Development
REST	Representational State Transfer
S3	Simple Storage Service (Object Storage)
SDLC	Software Development Life Cycle
SPARQL	
SSL/TLS	Secure Sockets Layer / Transport Layer Security

UI	User Interface
UX	User Experience
VM	Virtual Machine
NoSQL	Non-Relational Structured Query Language
Neo4j	Graph Database used for entity-relationship modeling
LoE	Level of Effort
CoT	Chain of Thought (prompt reasoning method)
PRINCE2	Projects in Controlled Environments, Version 2 (Project Management Methodology)
React	JavaScript library for building front-end interfaces
OCR	Optical Character Recognition
MongoDB	NoSQL Database for document storage
S3/MinIO	Object Storage for documents and OCR outputs
PDF	Portable Document Format
RDF/OWL	Resource Description Framework / Web Ontology Language
ELI	European Legislation Identifier
QA	Quality Assurance
JSONL	JSON Lines (for structured datasets)
CSV	Comma-Separated Values (data file format)
I/O	Input/Output

1. INTRODUCTION

1.1 Background & Literature Survey

1.1.1 Research Background

The proposed research focuses on creating an integrated, user-centered AI system for property law with the goal of bridging the gap between complex legal documents and public understanding. This study fills crucial gaps in existing legal AI systems, which often provide single-perspective summaries with no traceability mechanisms and produce erroneous visualizations that lack contextual awareness of legal relationships [1]. The total system comprises several modules, including legal document intake, multi-perspective summarization, risk and bias identification, knowledge graph generation, and visualization. These modules will work together to provide simplified but trustworthy interpretations of property law documents, allowing non-expert users including property owners, buyers, and citizens to better understand their rights, duties, and dangers. By integrating summarization, explainability, and visual analytics, the system attempts to make property law more transparent, accessible, and usable for decision-making [1] [2].

Legal reasoning in the realm of Sri Lankan property law - especially in interpreting deeds, leases, partition agreements, and title documents - necessitates far more than basic text retrieval or summarization. The complexity is rooted in a fragmented legal system shaped by colonial legacies, shifting statutory regimes, and localized land-use practices. These documents entwine overlapping legal norms, administrative practices, and socio-political histories - underscoring the need for AI systems that can replicate structured, statute- and precedent-based reasoning [1] [2] .

At present, large language models (LLMs) excel at language understanding but falter in high-stakes legal domains due to their lack of transparency and traceability. This is especially problematic in Sri Lanka, where property disputes are both pervasive and economically significant: unclear titles, co-ownership conflicts, and administrative delays undermine land's economic value and stifle access to credit [3] .

The demand for Explainable Artificial Intelligence (XAI) is clear: legal tools must do more than produce outcomes - they must articulate reasoning that aligns with the rule of law. In Sri Lanka's

fragmented land administration system - marked by squatting on state lands, inconsistent registration practices, and ethnically skewed colonization - explainability is essential for building trust and institutional legitimacy [4] [5] [6].

Techniques such as prompt chaining (chain-of-thought prompting) allow LLMs to articulate intermediate reasoning steps, improving interpretability [7] . Argumentation frameworks, like context graphs and probabilistic epistemic argumentation, help visualize these reasoning pathways - clarifying how statutes, precedents, and facts interconnect [8] [9] . Meanwhile, neuro-symbolic approaches - which combine LLMs with rule-based systems - offer both linguistic depth and legally sound logic structures [10] [11] .

However, there is a notable absence of such explainable AI applications specifically tuned to Sri Lankan property law, a system characterized by unique historical ordinances like the State Lands Ordinance, Land Development Ordinance, and the Registration of Title Act, as well as post-conflict land disputes in the North and East [12] [13] . A domain-specific reasoning system is urgently needed—one that can ingest Sri Lankan legal texts and documents, interpret them through statutory and precedent lenses, and produce transparent, auditable explanations. Such a system would enhance trust among legal practitioners, improve judicial efficiency, and contribute meaningfully to dispute resolution in land-related cases.

1.2 Research Gap

The integration of artificial intelligence (AI) into the legal sector has shown significant potential globally. Advances in large language models (LLMs) and legal-specific natural language processing (NLP) frameworks have facilitated tasks such as contract review, statutory analysis, and case summarization [14]. Despite these advancements, critical limitations remain, particularly when addressing reasoning transparency and jurisdiction-specific complexities, such as those inherent in Sri Lankan property law.

1.2.1 Lack of Explainability in Legal AI

Most existing legal AI systems are black boxes, producing outputs that cannot be explained or audited effectively [15] [16]. Although methods like chain-of-thought prompting and explainable

AI (XAI) frameworks have emerged [7], their application in legal contexts remains underexplored, and where implemented, they lack alignment with jurisdictional nuances. This opacity undermines trust and prevents adoption in high-stakes legal environments, where accountability and defensibility of reasoning are non-negotiable [6].

1.2.2 Global Focus, Limited Local Context

The majority of legal AI research and tools have been developed for jurisdictions like the United States, European Union, or China [17]. These systems are trained on datasets and legal corpora that do not reflect the unique statutory frameworks, legal instruments, and judicial precedents of Sri Lanka. Consequently, they fail to handle context-specific elements such as:

- The Registration of Documents Ordinance and its procedural nuances.
- Land tenure classifications under the Land Development Ordinance.
- Precedent-driven reasoning applied in property dispute litigation.

This jurisdictional mismatch creates a significant barrier to practical implementation in Sri Lanka.

1.2.3 Absence of Structured Legal Reasoning

While some global research efforts incorporate argumentation graphs and logic-based reasoning [8] [9], these remain largely experimental. There is no operational system that generates transparent, step-by-step legal reasoning chains linking facts to statutes and case precedents in a traceable manner. This leaves practitioners and stakeholders unable to validate AI outputs in complex property law scenarios, such as co-ownership disputes, encumbrance checks, or title verification.

1.2.4 Lack of Localized Datasets

Effective legal reasoning models rely on high-quality, annotated datasets. In Sri Lanka, digitized and structured legal corpora are fragmented and sparse. Statutes are often published in static PDFs, case law is scattered across archives, and deed registries remain predominantly manual [18] [12]. This scarcity of machine-readable, well-annotated data has hindered research and delayed the development of advanced, localized legal AI systems.

1.2.5 Unavailability of End-to-End Explainable Systems

Currently, no end-to-end explainable legal reasoning system exists for Sri Lanka that can:

- Ingest unstructured property documents.
- Cross-reference them against statutory provisions and precedents.
- Generate auditable, human-readable reasoning chains.

This lack of comprehensive solutions perpetuates reliance on manual processes, slowing dispute resolution and limiting access to justice.

The gap lies in the intersection of explainability, localization, and structured reasoning:

- Global legal AI systems offer capabilities without transparency.
- Emerging XAI frameworks lack jurisdiction-specific adaptation.
- Sri Lanka lacks digitized, interconnected datasets that enable reasoning.
- There is no operational system combining reasoning, explainability, and localization for property law.

Bridging this gap will require building an explainable reasoning module that integrates prompt chaining, argumentation graphs, and neuro-symbolic reasoning, trained and optimized for the statutory and precedent landscape of Sri Lankan property law. Such a system would not only close a critical research gap but also establish a foundation for trustworthy, auditable, and practical AI-driven legal assistance.

System / Feature	Explainability	Local Context	Structured Reasoning	Localized Datasets	End-to-End System
Current Global Legal AI Tools	X	X	X	X	X
Experimental XAI Frameworks	Partial	X	Partial	X	X
Static Legal Repositories (PDF)	X	Partial	X	Partial	X

Statutes, Case Archives)					
Academic Research Prototypes	Partial	✗	Partial	✗	✗
Proposed Explainable Reasoning Module	✓	✓	✓	✓	✓ (integrated, auditable)

Table 1 - Research Gap

1.3 Research Problem

1.3.1 Problem Discussion

Sri Lanka’s property law ecosystem is characterized by complex statutory frameworks, fragmented case law, and manual documentation processes that slow down legal reasoning and decision-making. Although AI and LLM-based systems have demonstrated remarkable potential for automating legal tasks globally [14] [16], they remain black boxes in high-stakes scenarios. This lack of explainability and transparency makes them unsuitable for contexts where traceable, auditable reasoning is required for dispute resolution, compliance, or judicial support.

Furthermore, there is no Sri Lanka-specific explainable reasoning module that integrates statutes, precedents, and factual inputs into structured, verifiable reasoning chains. Legal professionals, registrars, and policymakers lack tools that deliver context-aware, auditable insights, limiting the adoption of AI in critical workflows like title verification, partition actions, or encumbrance assessments [19] [12].

1.3.2 Research Questions

1. How can large language models be adapted to generate step-by-step, legally traceable reasoning chains for Sri Lankan property law?
2. What techniques—such as prompt chaining or argumentation graphs—are most effective for enhancing reasoning transparency in legal AI systems?
3. How can explainability improve trust, adoption, and accuracy in AI-assisted property law workflows?

1.3.3 Research Problem Statement

Current legal AI systems lack localized, explainable reasoning capabilities to produce transparent, auditable, and statute-aware outputs for Sri Lankan property law.

2. OBJECTIVES

2.1 Main Objective

To design and develop an explainable legal reasoning module for Sri Lankan property law that integrates large language models (LLMs), prompt-chaining techniques, and argumentation graphs to produce transparent, auditable, and statute-aware reasoning chains, thereby enhancing trust, interpretability, and adoption of AI-assisted legal tools in high-stakes workflows such as dispute resolution, title verification, and property transaction validation.

2.2 Specific Objectives

1. Develop a localized legal corpus
 - Compile and preprocess Sri Lanka-specific legal texts, including statutes, gazettes, judicial precedents, and annotated deeds.
 - Ensure the corpus is machine-readable.
2. Adapt and fine-tune LLMs for legal reasoning
 - Customize existing LLMs (e.g., GPT-4, LLaMA, LegalBERT) with Sri Lankan property law datasets.
 - Optimize for statute citation accuracy, legal terminology handling, and context-aware reasoning.
3. Integrate explainability through prompt chaining
 - Implement chain-of-thought prompting to generate step-by-step reasoning chains.
 - Enable the module to provide clear justification for every conclusion by linking reasoning steps to specific statutory provisions or case precedents.
4. Design and implement argumentation and logic graphs
 - Build visual argumentation frameworks to represent legal reasoning flows, conflicts, and evidence support.
 - Support interactive exploration of the reasoning process to enhance user trust and auditability.
5. Validate reasoning accuracy and explainability

- Collaborate with legal practitioners to assess the module’s accuracy, clarity, and usability.
 - Use quantitative metrics (e.g., citation accuracy, precision, recall) and qualitative expert feedback to refine the model.
6. Deploy a user-friendly reasoning interface
- Integrate the reasoning module into a prototype legal platform for property law professionals.
 - Ensure the interface allows querying, visualization, and export of reasoning results for legal documentation and case preparation.
7. Evaluate ethical, legal, and professional compliance
- Ensure the module adheres to data privacy regulations and AI ethics guidelines, including transparency and fairness principles outlined in Sri Lanka’s draft National AI Strategy.
 - Incorporate safeguards to flag low-confidence predictions for human review.

2.3 Outcome of the Objectives

By meeting these objectives, the project will deliver an end-to-end explainable reasoning engine that:

- Reduces ambiguity in AI-driven legal recommendations.
- Enhances trust and adoption among lawyers, registrars, and judicial officers.
- Provides a transparent, audit-ready foundation for AI systems supporting property law compliance, risk assessment, and dispute resolution in Sri Lanka.

3. METHODOLOGY

3.1 Development Methodology

The Agile development technique has been used for this project in order to facilitate experimentation, modular implementation, and ongoing enhancements in line with research on AI-driven legal reasoning. Agile's sprint-based methodology offers adaptability to changing design choices, integration needs, and legal professionals' professional input. A particular job, like data preprocessing, model fine-tuning, API integration, or front-end visualization, will be the focus of each sprint, allowing for incremental progress, validation, and course correction as necessary.

Because of its versatility and appropriateness for research-oriented projects including interdisciplinary collaboration, technical experimentation, and iterative revisions, this methodology is chosen over more conventional models like Waterfall. In order to keep the reasoning module in line with research objectives, technical viability, and the practical demands of legal practitioners, the Agile process facilitates a feedback-driven and flexible development cycle.

Development Methodologies Comparison

Methodology	Description
Waterfall Model	A phased, linear process in which every step must be finished before going on to the next. Although it is well-organized and simple to administer, it is not adaptable enough to take into account fresh discoveries or modifications made during continuing study. Unsuitable for AI initiatives that need ongoing testing and improvement.
RAD (Rapid Application Development)	Focuses on quick feedback cycles and rapid prototyping. It is less feasible for research projects with limited stakeholder availability, nevertheless, due to its high resource requirements and requirement for constant user interaction.
Prototyping Model	Focuses on developing early system iterations for stakeholder review and input. Although useful for improving user-facing features, it might not be

	entirely compatible with model-centric, back-end-heavy research that places a higher priority on correctness of reasoning and performance.
Agile Model	For research that needs incremental development, model testing, and prompt-driven refinement, this adaptable and iterative technique is perfect. promotes constant feedback and quick adjustment to new information. Ideal for AI projects with constantly evolving stages such as evaluation, model fine-tuning, and data preprocessing.

Table 2 - Development Methodologies Comparison

The Agile model was chosen because of the project's experimental nature and the requirement for ongoing validation across data, reasoning models, and evaluation outcomes. It is the best method for effectively developing and improving the explainable reasoning module while staying in line with the project goals and knowledge graph integration requirements because it supports modular design, iterative development, and stakeholder-driven feedback.

3.2 Project Management Methodology

The Agile PRINCE2 methodology was used for this research-driven project's efficient planning, implementation, and monitoring. This hybrid paradigm is ideal for an academic research project that involves changing technological goals, ongoing experimentation, and interdisciplinary collaboration because it blends the iterative flexibility of Agile with the structured governance and documentation rigor of PRINCE2.

Agile PRINCE2 facilitates gradual delivery, regular validation with supervisors and legal experts, and stage-based planning that is in line with academic milestones. This guarantees that the project keeps a balance between risk management, academic discipline, and flexibility to take into account new information as it is being developed and tested.

Justification for Agile PRINCE2

Methodology	Description	Suitability
PRINCE2	A methodical, process-oriented approach that emphasizes milestone-based planning, risk management, and documentation.	Strong governance is offered, but it lacks the flexibility required for experimental AI research.
Agile PM	Focuses on flexibility, stakeholder input, and quick iteration, but it has less formal structure.	Extremely flexible, but may find it difficult to manage academic milestones and the need for thorough documentation.
Agile PRINCE2 (Selected)	A hybrid strategy that combines the reactivity of Agile with the structure of PRINCE2. It is perfect for this project since it provides formal project controls	It is perfect for this project since it provides formal project controls while encouraging iterative experimentation and ongoing improvement.

Table 3 - Justification for Agile PRINCE2

3.3 Scope Management

To guarantee focus, viability, and alignment with the research goals, the project scope has been precisely defined:

3.3.1 In-Scope Activities

- Gathering and preprocessing information about legal texts unique to Sri Lanka, including as statutes, court rulings, and administrative records.
- Improving the explainability and reasoning of big language models in property law contexts.
- The creation of front-end interfaces for querying and visualization using React.js and back-end services based on FastAPI.
- Integration with the knowledge graph module to improve traceability and reasoning accuracy.

- Analyzing reasoning outputs using expert assessments and quantitative metrics.

3.3.2 Out-of-Scope Activities

- The system's commercial rollout or production-grade deployment.
- The use of private or sensitive legal documents because of privacy and ethical restrictions.
- Complete automation of legal decision-making; human supervision will always be necessary.

3.3.3 Milestone Planning and Schedule

The project will be organized into sprints and phases, each of which will correspond to technical and scholarly deliverables:

Deliverable / Milestone	Planned Date
Topic exploration and brainstorming	July 1 – July 10, 2025
Project topic finalized	July 15, 2025
Topic Assessment Form (TAF) submission	July 20, 2025
Project proposal finalized	August 25, 2025
Literature review drafted and approved	September 15, 2025
System architecture design and tool selection	October 10, 2025
Data collection and preprocessing completed	October 31, 2025
Initial LLM fine-tuning and reasoning module prototype	November 25, 2025
Integration with knowledge graph and front-end UI	December 15, 2025
Testing, debugging, and evaluation completed	January 10, 2026
Make Improvements	January 20, 2026
Final report writing and documentation	January 31, 2026
Final submission and presentation	February 5, 2026

Table 4 - Milestone Planning and Schedule

A detailed Gantt chart outlining these milestones is provided at [Appendix-III](#).

3.3.4 Deliverables

- Datasets of Sri Lankan property law documents that are organized and clean.
- A useful reasoning API built on FastAPI that can process queries and produce clear reasoning chains.
- A React.js interactive interface for displaying legal logic graphs and reasoning procedures.
- Evaluation reports that include results of expert validation, explainability, and correctness.
- Detailed documentation of the architecture, design, and assessment procedures of the system.

3.4 Implementation Strategy

The implementation of the Explainable Legal Reasoning Module will follow a structured, multi-phase approach to ensure that data collection, model training, system integration, and deployment are all aligned with the project's objectives. The strategy integrates state-of-the-art AI practices, modular development techniques, and continuous validation with domain experts to deliver a robust, explainable, and user-centric reasoning system.

3.4.1 Phase 1 – Data Collection and Preparation

To achieve reasoning accuracy and localization, the first step involves creating a comprehensive, machine-readable corpus of Sri Lankan property law. Data source details are added to [Appendix-I](#).

Data Sources

- Primary legal texts: Statutes, ordinances, and regulations such as the Registration of Documents Ordinance, Land Development Ordinance, and Title Registration Act.
- Case law: Judgments from the Supreme Court and Court of Appeal related to property disputes.
- Gazettes and circulars: Notices of land acquisition, government grants, and regulatory updates.
- Reference materials: Legal commentaries, manuals, and academic publications for context enrichment.

Data Preprocessing

- OCR and text extraction from scanned legal documents.
- Cleaning and normalization to remove noise, standardize terminology, and segment documents into structured sections (e.g., clauses, definitions, references).
- Annotation: Tagging statutes, case references, and procedural steps to support reasoning workflows.

3.4.2 Phase 2 – Model Training and Fine-Tuning

The reasoning module will be powered by a fine-tuned large language model (LLM) optimized for Sri Lankan property law.

Model Selection and Preparation

- Base models: GPT-4, LLaMA, or LegalBERT for legal text understanding.
- Parameter-efficient tuning methods such as LoRA or QLoRA to adapt models without requiring high compute resources.

Training Process

1. Dataset preparation: Curated corpus split into training, validation, and evaluation sets.
2. Fine-tuning: Training the model for tasks such as statute identification, clause interpretation, and reasoning-chain generation.
3. Prompt engineering: Creating structured prompts to guide chain-of-thought reasoning for property law scenarios.
4. Evaluation: Using metrics like accuracy, citation recall, and expert-rated reasoning clarity.

3.4.3 Phase 3 – Backend Development (FastAPI)

The backend forms the core processing engine for reasoning tasks:

- Inference API: Endpoints to handle queries, retrieve reasoning chains, and return results in JSON for front-end consumption.

- Integration with MongoDB: Storing user inputs, processed documents, reasoning steps, and model outputs for auditing and future improvement.
- Logging and Monitoring: Capturing usage metrics and errors to ensure reliability and traceability.
- Security: API-level authentication and authorization to protect sensitive legal queries and datasets.

3.4.4 Phase 4 – Front-End Development (React)

The React-based front-end will provide a user-friendly, interactive interface designed for legal professionals and researchers:

- Query Interface: Allows users to submit legal questions or upload documents.
- Visualization Panel: Displays reasoning chains in textual and graph-based formats using libraries like D3.js or Plotly.
- Interactive Controls: Options to explore reasoning steps, review supporting statutes, and export explanations for documentation.
- Responsive Design: Optimized for desktops and tablets to support varied user environments.

3.4.5 Phase 5 – Database Layer (MongoDB)

MongoDB will serve as a flexible and scalable NoSQL database to store:

- User queries and metadata.
- Document embeddings and reasoning chains.
- System logs and evaluation data for monitoring and continuous learning.

Key Features:

- Schema-less structure for diverse legal data types.
- Horizontal scaling to support increased usage over time.
- Integrated indexing for faster retrieval of historical reasoning sessions.

3.4.6 Phase 6 – Integration with Knowledge Graph

To enhance reasoning accuracy and explainability, the module will integrate with Sharan's Knowledge Graph component:

- Cross-referencing statutes, case law, and procedural documents during reasoning.
- Highlighting relevant entities and relationships to justify model outputs.
- Leveraging the KG to update reasoning workflows dynamically when legal updates are ingested.

3.4.7 Phase 7 – Testing and Validation

A multi-layered testing strategy will be implemented to ensure accuracy, stability, and usability:

- Unit Testing: API endpoints, database functions, and front-end components.
- Integration Testing: End-to-end workflows from document upload to reasoning output.
- Expert Validation: Periodic reviews by legal professionals to assess reasoning quality, statute citation accuracy, and clarity of explanations.
- Accuracy: LLM-AS-JUDGE method will be experimented.
- Performance Metrics:
 - Technical: Response time, API reliability.
 - Legal: Citation precision, reasoning transparency, and compliance with statutes.

3.4.8 Phase 8 – Deployment and Monitoring

- Containerization: Using Docker for reproducible and scalable deployments.
- Cloud Hosting: Hosting APIs and database in a secure cloud environment with regular backups.
- Monitoring: Real-time logs and performance dashboards for proactive issue detection.
- Continuous Improvement: Feedback-driven updates to prompts, model weights, and UI components.

3.4.9 Continuous Learning and Updates

The reasoning module will incorporate a feedback loop:

- Capturing user interactions and flagged errors.
- Using these insights for incremental model retraining and prompt refinement.
- Regular updates to align with new statutes, legal amendments, or precedents ingested by the knowledge graph.

3.5 System Architecture

The overall architecture of the LegalVision platform follows a layered design, ensuring modularity, scalability, and clear separation of concerns between user interaction, application logic, data management, and external integrations.

A detailed system architecture diagram is provided in [Appendix-II](#). The platform is divided into four primary layers, described below:

3.5.1 Presentation Layer (React)

The **front-end interface**, built with React, enables:

- Submission of legal queries and property documents.
- Visualization of reasoning steps, chains, and logic graphs.
- Exploration of knowledge graph entities and relationships.
- A responsive and intuitive experience optimized for legal professionals and academic users.

3.5.2 Application Layer (FastAPI)

This layer serves as the **core logic layer** and orchestrates:

- API Gateway: Handles authentication, authorization, and request routing.
- Reasoning Module: Processes user queries, triggers prompt chaining and chain-of-thought reasoning, and integrates outputs from the LLM.

- Knowledge Graph Module: Executes entity and relationship queries to provide statute references, case connections, and contextual data to the reasoning engine.

3.5.3 Data Layer (MongoDB & Neo4j)

This layer manages **data storage and retrieval**:

- MongoDB: Stores user queries, reasoning chains, usage logs, and evaluation records.
- Neo4j GraphDB: Stores statutes, cases, entities, and relationships for knowledge graph queries.
- Object Storage (S3 or Local): Stores uploaded documents, OCR outputs, and processed files for traceability and future training use.

3.5.4 External Integration Layer

This layer connects the system to external resources:

- Legal Data Sources: Automated ingestion of gazettes, statutes, and case law for continuous updates.
- LLM Providers: Secure access to hosted LLM APIs (e.g., GPT-4) or self-hosted models for inference and reasoning tasks.

This layered approach ensures a scalable and maintainable architecture, allowing independent updates to the front-end, back-end, and data modules while maintaining robust integration for reasoning and knowledge graph functionalities.

3.6 Context Diagram

The high-level interactions between the key elements of the Knowledge-Driven AI Framework for Legal Contract Understanding and Trust Assessment are depicted in the context diagram. In order to analyze documents, reason through legal content, create dynamic knowledge graphs, and assess bias and risk, this framework combines four essential modules that function together effortlessly.

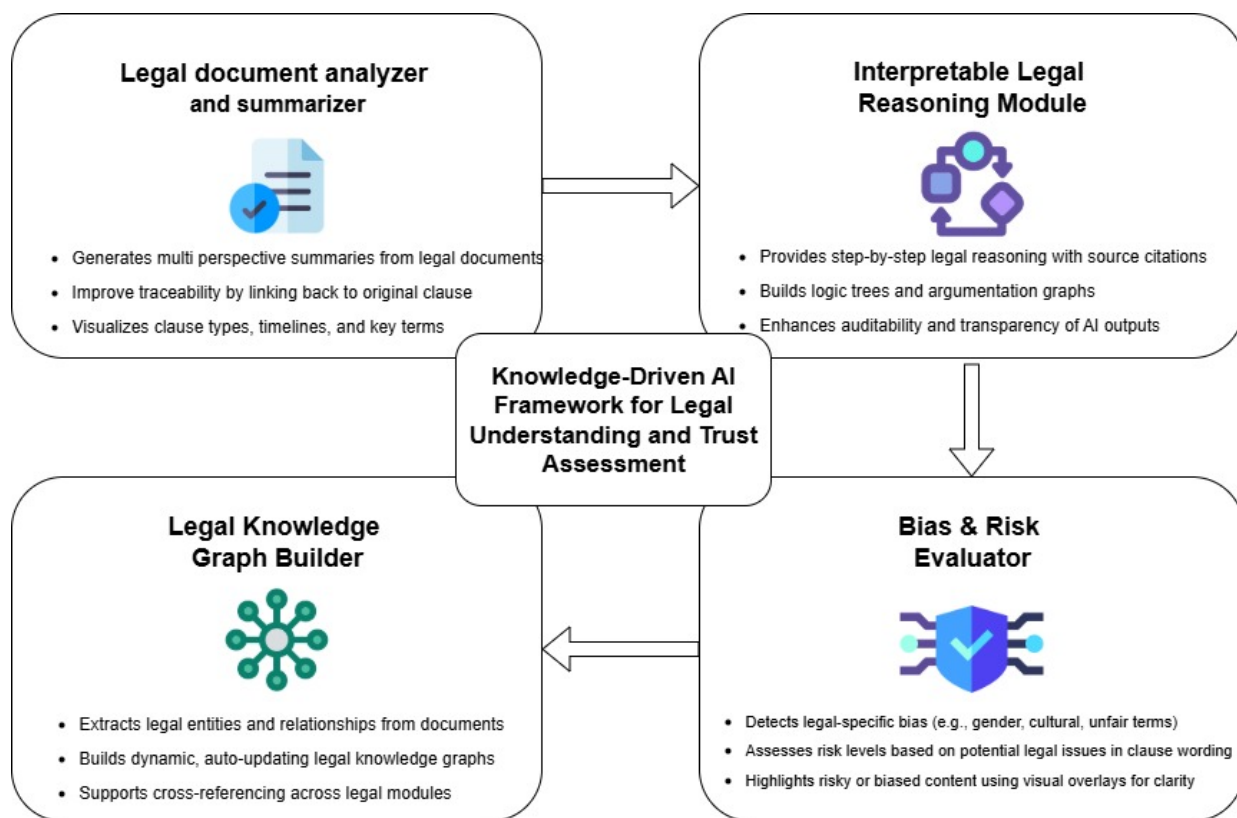


Figure 1 - Context Diagram

3.7 Ethical and Legal Compliance

To guarantee data privacy, security, and equity, this initiative closely adheres to legal and ethical requirements. In accordance with Sri Lanka's Personal Data Protection Act, No. 9 of 2022, all legal papers and user inputs are handled. All data is protected via encrypted databases and role-based access controls, and sensitive information is anonymized before being stored. The system is kept transparent and auditable by logging each query and reasoning output for traceability.

The models are trained on a variety of legal datasets and are regularly checked for potential biases, such as gender- or cultural-related skew, in order to preserve equity and confidence. Clear reasoning steps with source citations are included in the outputs so that users may comprehend the decision-making process. In order to ensure ethical and responsible use throughout its development and implementation, the system is intended to be a helpful legal instrument rather than a substitute for expert judgment.

3.8 Risk Management

To guarantee the system's stability, dependability, and ethical compliance, it is essential to identify possible risks and put appropriate mitigation techniques into place. The main dangers, their seriousness, and the tactics used to lessen their effects are compiled in the table below.

Risk	Severity	Mitigation Strategy
Outdated or incomplete legal information	High	Plan frequent data audits and set up automatic data ingestion pipelines from official sources (registries, case law portals, and gazettes).
Inaccurate reasoning outputs or model bias	High	Incorporate a bias detection module, use a variety of training datasets, and incorporate reviews from legal experts into the validation process.
Breach of data privacy	High	Encrypt transmission and storage, anonymize sensitive data, and use role-based permissions and authentication to limit system access.
Module integration issues	Medium	Use rollback mechanisms to recover from errors, establish error reporting, and carry out comprehensive integration testing.
Problems with scalability or performance	Medium	Use containerized deployments for simple scalability, optimize database and API queries, and keep an eye on system load to schedule upgrades.
Misunderstanding of outputs by the user Moderate	Medium	Incorporate explicit disclaimers into the user interface, make the logic transparent through citations, and support expert evaluation of the findings.
Delays in the project timeframe	Low	Make proactive schedule adjustments based on milestone tracking and use Agile sprints with frequent progress assessments.

Limitations in hardware resources	Low	For model hosting and backups, use cloud-based solutions to guarantee high availability and alternatives for recovery.
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Table 5 - Risk Management

4. PROJECT REQUIREMENTS

4.1 Functional Requirements

The functional requirements define the core features and capabilities that the reasoning module must deliver for reliable and explainable performance.

ID	Requirement	Description
FR-01	Legal Data Ingestion	Collect and preprocess statutes, gazettes, case law, and property-related documents into structured, machine-readable formats suitable for reasoning.
FR-02	Reasoning Chain Generation	Generate step-by-step reasoning chains using LLMs with prompt chaining and logic-based templates to ensure interpretability and traceability.
FR-03	Argumentation Graphs	Produce visual reasoning graphs that link facts, statutes, and case precedents to improve explainability for legal professionals and non-technical users.
FR-04	Case-Specific Analysis	Support the analysis of uploaded property-related documents (e.g., deeds, partition orders) and map them to relevant legal rules and precedents.
FR-05	Integration with Knowledge Graph	Cross-reference legal entities and relationships from the Knowledge Graph for context-aware reasoning and enriched outputs.
FR-06	Audit Trail	Store reasoning steps, source references, and user interactions to support audits and future validation.
FR-07	API Accessibility	Provide REST/GraphQL APIs for seamless integration with external systems, front-end interfaces, and other LegalVision modules.
FR-08	Interactive Visualization	Enable users to explore reasoning chains interactively, drilling down into supporting statutes and references.

FR-09	Export and Reporting	Allow export of reasoning chains and visual graphs in formats such as PDF, JSON, or CSV for documentation and evidence submission.
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Table 6 - Functional Requirements

4.2 Non-Functional Requirements

These requirements define how the system should operate to ensure reliability, security, and usability in high-stakes legal environments.

ID	Requirement	Description
NFR-01	Performance	Generate reasoning chains and visual graphs within 3–5 seconds for typical queries, ensuring efficiency during live legal consultations.
NFR-02	Scalability	Support concurrent usage by 50+ users with growing datasets, without degrading performance or accuracy.
NFR-03	Security	Enforce strong API authentication, encrypted storage, and access control to comply with the Sri Lanka Personal Data Protection Act (No. 9 of 2022).
NFR-04	Reliability	Maintain a 99% uptime, with automatic backup and disaster recovery protocols to prevent data loss.
NFR-05	Auditability	Log every query, reasoning chain, and user interaction to ensure transparent audit trails for compliance and error analysis.
NFR-06	Usability	Provide an intuitive front-end interface with clear navigation and visual aids suitable for both legal experts and non-technical users.
NFR-07	Maintainability	Use a modular code architecture to allow updates or enhancements without disrupting ongoing operations.
NFR-08	Integration Compatibility	Ensure smooth interoperability with the Knowledge Graph module and external legal systems through REST, GraphQL, and standard data formats.

NFR-09	Accuracy Monitoring	Continuously evaluate outputs with predefined metrics (e.g., citation precision, reasoning accuracy) and flag low-confidence outputs for review.
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Table 7 - Non-Functional Requirements

4.3 Expected Test Cases

Test Case ID	Scenario	Expected Result
TC-01	Data Ingestion – Statutes Upload a statute PDF into the system.	Text is extracted, normalized, and stored in a machine-readable format, ready for reasoning queries.
TC-02	Data Ingestion – Case Law Integrate a case judgment.	Case entities and references are correctly identified and linked to relevant statutes and precedents.
TC-03	Property Document Analysis Upload a deed or partition order.	The system identifies relevant clauses, statutes, and potential conflicts, generating a clear reasoning base.
TC-04	Reasoning Chain Generation Submit a query about a title dispute.	The system generates a step-by-step reasoning chain referencing statutes and case law, with supporting explanations.
TC-05	Argumentation Graph Visualization View reasoning output for a completed query.	An interactive graph displays facts, statutes, and relationships in a clear, auditable structure.
TC-06	Knowledge Graph Integration Run a query requiring data cross-referencing.	Output integrates context and entities from the Knowledge Graph, improving accuracy and relevance.

TC-07	API Query Call reasoning API via REST or GraphQL.	Returns accurate JSON-formatted results within performance benchmarks.
TC-08	Audit Logging Review logs after a session.	All queries, reasoning steps, and outputs are recorded with timestamps and user identifiers for traceability.
TC-09	Performance Stress Test Simulate 50+ concurrent users.	System maintains response times under 5 seconds, without failures or data loss.
TC-10	Backup and Recovery Simulate a failure during reasoning execution.	System restores previous state without data corruption or loss of audit logs.
TC-11	Accuracy Monitoring Analyze flagged low-confidence outputs.	Flagged reasoning chains are highlighted, and the system recommends manual review.
TC-12	Export and Reporting Export reasoning output for a case.	PDF or JSON export includes the reasoning chain, references, and visual graph snapshots.
TC-13	Security Check Attempt unauthorized API access.	Access is denied, the attempt is logged, and no data exposure occurs.

Table 8 - Expected Test Cases

DESCRIPTION OF PERSONAL AND FACILITIES

Our project team is made up of experts in artificial intelligence, natural language processing, and software engineering, and each brings a distinct skill set to the creation of the explainable legal reasoning module. AI and data science experts will work on fine-tuning large language models (e.g., GPT, LegalBERT) with Sri Lankan property law datasets, while others will handle system architecture design, backend services using FastAPI, and frontend development with React for reasoning visualization. Furthermore, database skills will help with the integration of MongoDB for document storage and Neo4j for knowledge graph creation.

To aid this task, we will use current computer facilities outfitted with high-performance GPUs for model training and evaluation, ensuring efficient processing of complicated legal documents. Cloud-based development environments, version control systems, and collaborative repositories will allow for efficient teamwork and iterative improvements. Dedicated testing environments will be created to imitate real-world legal issues, allowing us to check reasoning accuracy, explainability, and system stability prior to full-scale implementation.

COMMERCIALIZATION

The proposed system demonstrates strong commercialization potential within the legal and regulatory sectors, with primary markets including law firms, notary offices, government land registries, and academic institutions engaged in legal research. These stakeholders face mounting challenges in managing property-related disputes and documentation efficiently, and the system addresses these challenges by offering a transparent, explainable, and trustworthy AI-driven solution.

For legal professionals, the system delivers several key benefits:

- **Transparent Reasoning:** Generates auditable reasoning chains directly linked to statutes and case precedents, reducing dependency on manual interpretation.
- **Efficiency Gains:** Automates property document analysis and title verification, significantly reducing time spent on repetitive legal tasks.
- **Fairness and Accountability:** Provides explainable outputs that enhance trust and confidence in AI-assisted legal processes.

From a cost–benefit perspective, organizations stand to gain by reducing dispute resolution delays, minimizing administrative workload, and improving decision-making accuracy. Furthermore, the system’s modular and scalable architecture enables future enhancements such as integration with e-Government portals and advanced analytics dashboards. These extensions will ensure the platform remains adaptable, relevant, and competitive.

By delivering measurable efficiency improvements and reinforcing fairness in legal workflows, the system is well-positioned to become a trusted, AI-driven legal reasoning solution that can modernize and transform Sri Lanka’s property law ecosystem.

BUDGET AND BUDGET JUSTIFICATION

Component	Cost (LKR)	Justification
Travel & Meetings with Law College Experts	10,000	To seek legal advice and acquire datasets. Includes transportation, refreshments, and coordination (around 2-3 trips).
Data Collection	5,000	Purchase or gain access to legal datasets essential for analysis and model training (such as property law case reports and registries).
Research Paper Publication	100,000	Covers the cost of publishing in reputable journals and conferences.
Internet Charges	24,000	Continuous internet connectivity (2,000 LKR × 12 months) is needed for dataset downloads, literature surveys, virtual meetings, and cloud-based experiments.

Table 9 – Budget and Budget Justification

Total Estimated Budget: 139,000 LKR

The suggested funding fulfills the research's key needs. Travel and meeting costs make it easier to interact with legal professionals and get datasets. Data collection guarantees that trainees and

analysts have access to reliable legal texts such as property law records and case reports. Publication fees allow for the dissemination of discoveries in reputable journals and conferences, assuring academic significance. Internet charges ensure continuous access to dataset downloads, virtual collaboration, and cloud-based experimentation. Overall, the funding is balanced, allowing for both successful research execution and impactful dissemination.

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APPENDICES

Appendix-I Resources for Sri Lankan Property Law

1. Official Legislation Portals

- SriLankaLaw Revised Statutes – Offers complete texts of key property laws such as the Registration of Documents Ordinance (Cap. 300), outlining definitions, registration procedures, and legal effects of deeds, leases, and mortgage instruments [LawGratis+2ResearchGate+2ResearchGate+4Sri Lanka Law+4FAOLEX+4](#).
- FAOlex / Government PDF – Provides the full text of the Land Development Ordinance (No. 19 of 1935), governing the development and alienation of state land [Wikipedia+4FAOLEX+4Land Portal+4](#).
- Wikipedia Summaries:
 - Land Development Ordinance – Summarizes history and provisions enabling state land resettlement [SearchWorks+9Wikipedia+9FAOLEX+9](#).
 - State Lands Ordinance (No. 8 of 1947) – Covers Crown land classification and governance mechanisms [Wikipedia+1](#).
- SriLankaLaw Articles – Example: Article on the Registration of Title Act, No. 21 of 1998, detailing its transformative role in property title clarification [Lanka Law+15Lanka Law+15resources.ials.sas.ac.uk+15](#).

2. Legal Research and Case Law

- vLex Sri Lanka Collection – A subscription-based platform offering a vast repository of Sri Lankan legal documents—consolidated Acts, Supreme Court Reports, New Law Reports, etc. [vLex](#).
- Bodleian Law Cases Guide – Lists open-access resources for Sri Lankan Supreme Court and Court of Appeal judgments (New Law Reports from 1878 onward) [libguides.bodleian.ox.ac.uk+1](#).

- GlobaLex (NYU Law) – An authoritative guide to Sri Lankan legal research, including colonial legal history, court system structures, and resource bibliographies [GlobaLex+2resources.ials.sas.ac.uk+2](#).

3. Legal Overviews & Context

- LawGratis “Property Law in Sri Lanka” – A blog entry (Mar 2025) that covers legal foundations such as Roman-Dutch and English Common Law influences, the Land Development Ordinance, Registration of Titles, and other property frameworks [LawGratis+1](#).
- LandPortal – Sri Lanka Narrative – Offers insights into land governance, including the LDO’s context and implications in settling state lands [Land Portal+1](#).

4. Government Gazette

- Sri Lanka Gazette – The official publication for statutory announcements, including enactment of new laws, land notifications, government proclamations, and legislative updates [Wikipedia](#).

5. Institutional & Library Resources

- Law & Society Trust (LST) – A premier Sri Lankan legal research institution offering access to Acts, Law Reports, Gazette archives, and legal periodicals (not directly cited here, but highly recommended).
- Survey Department & Ministry Resources – While not fully captured here, these entities publish land and cadastral data and may provide additional legal frameworks around land mapping and regulation (see Ministry of Urban Development context) [Wikipedia](#).

6. Additional Tools & Educational Resources

- Scribd Presentation – Provides educational summaries comparing the Registration of Documents Ordinance (1927) and the Registration of Title Act (1998), highlighting procedural evolution [ResearchGate+6Scribd+6Studocu+6](#).

Resource Summary Table

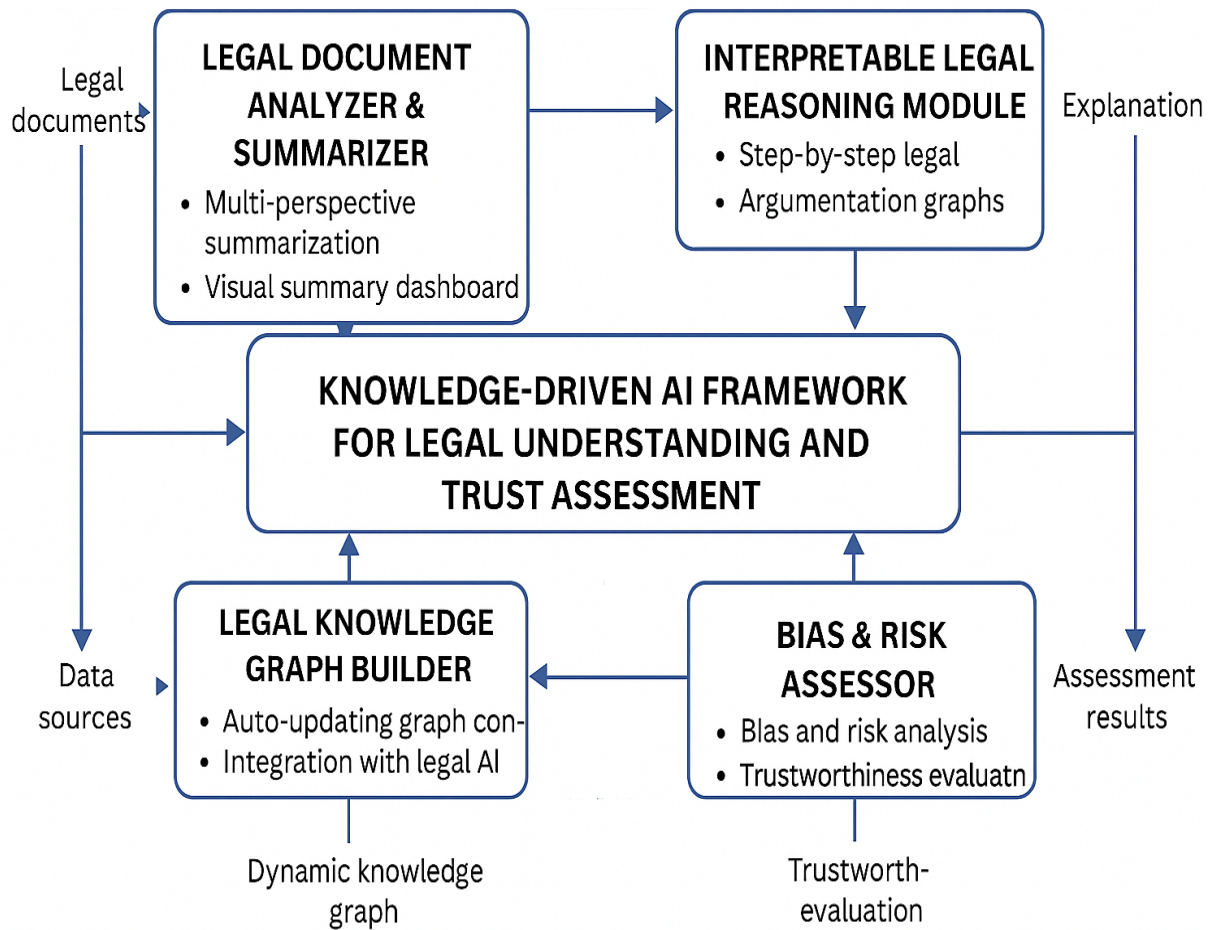
Resource / Platform	Description
SriLankaLaw Revised Statutes	Official texts for statutes like the Registration of Documents Act
FAO / Government PDFs	Access to full Land Development Ordinance texts
Wikipedia Summaries	Useful overviews of major land laws (with citations)
SriLankaLaw Articles	Insights into modern laws like title registration
vLex Sri Lanka	Paid platform with legislation and case law compilation
Bodleian Law Cases Guide	Free access guide to older Sri Lankan case law
GlobaLex (NYU Law)	Legal system overview and resource bibliography
LawGratis	Blog overview of property law frameworks
LandPortal Narrative	Governance context and land administration discussion
Sri Lanka Gazette	Primary source for new laws and notifications
Institutional Repositories (e.g., LST)	Deep archives of statutes, judgments, gazettes
Scribd Educational Materials	Visual comparisons of deed vs title systems

References for Data Sources

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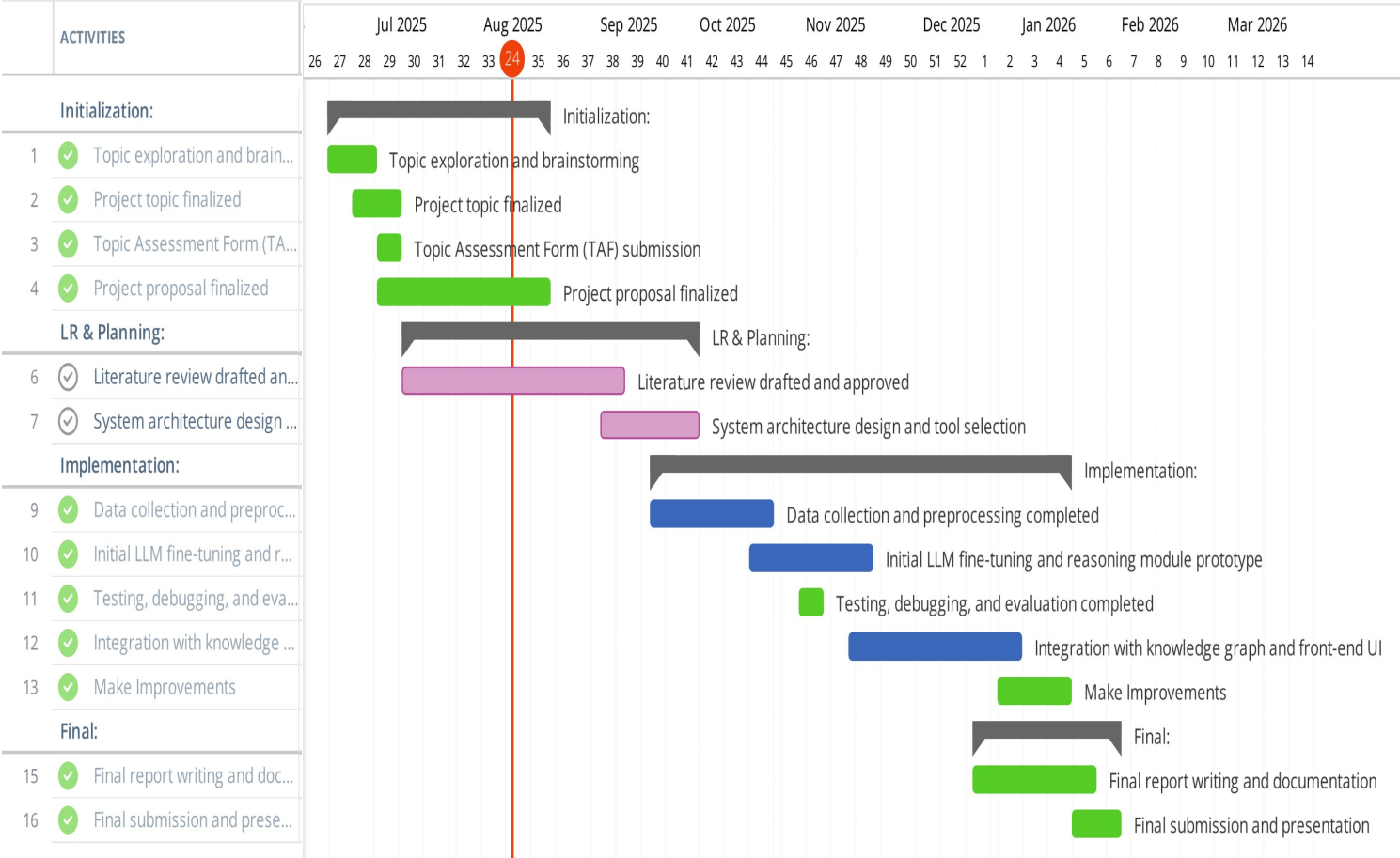
Appendix-II System Architecture Diagram



Appendix-III Gantt Chart

Feature Planning

Read-only view, generated on 24 Aug 2025



Work Breakdown Structure

